**Task 9 — Error Analysis (Retake / Improved)**

**Model:** Weighted RoBERTa-Large (Emotion Classification)
**Dataset:** TV transcript sentences (Australian Survivor style)
**Labels:** anger, disgust, fear, happiness, neutral, sadness, surprise

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**1. Introduction**

In this task, I analyzed the prediction results of our best model: **Weighted RoBERTa-Large**. The goal of error analysis is to understand what the model does well, where it fails, and what patterns explain the mistakes. This is important for the client use-case (emotion tagging in TV transcripts), because wrong predictions can lead to incorrect episode tagging and misleading content insights. Since the dataset is imbalanced, I focus on **macro F1-score** because it treats all emotions equally.

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**2. Overall Results**

The model achieved:

- **Accuracy:** 0.693
- **Macro F1:** 0.632
- **Weighted F1:** 0.692

**Interpretation:** Accuracy is influenced by frequent classes (especially neutral and happiness). Macro F1 is the fairest score for this task because it reflects performance across all emotions, including rare ones like disgust. The lower macro F1 compared to weighted F1 shows the model still struggles on rare or subtle emotions.

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**3. Confusion Patterns (Emotion overlaps)**

The confusion matrix shows that the model predicts **neutral** and **happiness** most accurately, but there are consistent overlaps:

- **Neutral → Happiness:** slightly positive neutral lines are predicted as happiness.
- **Fear → Neutral:** fear without strong emotion words is often predicted as neutral.
- **Sadness → Fear / Neutral / Happiness:** calmer sadness is sometimes missed or softened.
- **Surprise → Happiness / Neutral:** similar expressions and punctuation cause confusion.
- **Disgust:** weak performance because there are very few examples.

This makes sense in TV transcripts because emotions are often indirect and depend on context that is missing when classifying one sentence at a time.

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**4. Per-Class Performance (Strengths and weaknesses)**

- **Strongest:**
    - **Happiness (F1 ≈ 0.77)**
    - **Neutral (F1 ≈ 0.72)**
      These classes appear frequently and have clearer patterns.
- **Medium:**
    - **Fear (F1 ≈ 0.65)**
    - **Sadness (F1 ≈ 0.66)**
      These overlap in wording (worry/loss/uncertainty).
- **Weakest:**
    - **Anger (F1 ≈ 0.50)** because it overlaps with other high-intensity expressions and punctuation effects.
    - **Disgust (F1 ≈ 0.40)** because it is extremely rare in the data.

Overall, the model works best when emotion is clear and common, and struggles when emotion is rare, subtle, or overlaps with other emotions.

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**5. Most Common Confusions (Concrete counts)**

The most frequent confusion pairs were:

- **neutral → happiness (31)**
- **sadness → fear (15)**
- **fear → neutral (12)**
- **fear → anger (10)**
- **neutral → fear (10)**

**Meaning:** The model often over-reads emotion in neutral lines, especially happiness and fear. Sadness and fear share similar tone and vocabulary, making them hard to separate. Neutral is difficult because it can contain weak emotional cues that sound slightly positive or negative.

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**6. Concrete Error Examples (with explanation)**

| Sentence | True | Predicted | Why it failed |
|---|---|---|---|
| “It wasn't the only moral dilemma Tara faced.” | neutral | fear | The model links “dilemma” with fear even though the sentence is informational. |
| “I don't even feel bad.” | neutral | anger | Dismissive tone can look like anger without context. |

| Sentence | True | Predicted | Why it failed |
|---|---|---|---|
| “Tara sweet-talked the tribe into voting with her.” | neutral | disgust | Persuasion is interpreted as manipulation/negative intent. |
| “That's going to be his downfall.” | fear | anger | Strong wording interpreted as anger instead of fear/worry. |
| “Go!” | anger | happiness | Exclamation is read as excitement; too little context. |

These examples show the model sometimes reacts strongly to **keywords** or **punctuation** instead of full meaning.

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**7. Sentence Length vs Accuracy (more detailed + concrete examples)**

I checked sentence length to see if longer sentences cause more mistakes.

- **Correct predictions average:** ~8.4 words
- **Incorrect predictions average:** ~9.8 words

This suggests longer sentences are harder, but the key insight is *why* length affects performance:

**A) Very short sentences (low context)**
Short lines often have too little information, so the model guesses using punctuation or patterns.

- Example: **“Go!”**
  True: anger → Predicted: happiness
  Reason: without context, the “!” can look like excitement.

**B) Long sentences (mixed signals)**
Longer sentences often contain multiple clauses or mixed emotional cues (e.g., narration + emotion word). The model may focus on one word and ignore the overall meaning. This increases confusion between similar emotions (sadness vs fear) and causes neutral lines to be predicted as emotional.

**Conclusion:** The model is most reliable on **medium-length sentences with clear emotion words**, and less reliable on very short or very long sentences.

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**8. Common Words in Wrong Predictions**

The most common words in wrong predictions were neutral functional words (e.g., “to”, “I”, “you”, “the”, “and”). This shows many mistakes happen when sentences lack strong emotional words, so the model has weaker signal and relies on subtle cues.

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**9. Errors per True Emotion**

The model makes the most errors on **neutral, sadness, and fear**, which are often subtle and overlap in language. Disgust and surprise show fewer total errors mainly because they have fewer examples, not because the model is strong on them.

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**10. Improvements (based on the observed errors)**

To improve performance and make the model more reliable for transcript tagging:

- Add more examples for rare emotions (especially **disgust** and **surprise**).
- Include context (previous subtitle line) for very short sentences.
- For long sentences, split into clauses or predict at phrase-level.
- Use confidence thresholds and flag uncertain predictions for review.

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**11. Conclusion**

The Weighted RoBERTa-Large model performs well overall (**0.693 accuracy**, **0.632 macro F1**), especially for **neutral** and **happiness**. However, it struggles with rare emotions, subtle emotions, and cases where sentences are very short (no context) or long (mixed signals). This error analysis identifies the main failure patterns and suggests practical improvements for more reliable client-ready transcript tagging.